EVC-Net: A Hybrid Deep Learning Network for Breast Cancer Classification from Histopathological Images

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Breast cancer is a prevalent and life-threatening disease where early and accurate diagnosis is critical for effective treatment. Conventional histopathological analysis, while the standard for diagnosis, can be laborious and is subject to inter-observer variability, highlighting the need for robust automated methods. This article introduces EVC-Net, a novel hybrid deep learning framework designed to automate the classification of breast cancer from histopathological images. EVC-Net synergistically integrates an EfficientNetV2S for fine-grained texture feature extraction, a vision transformer (ViT) for capturing global context, and a capsule network to preserve spatial hierarchies within tissue structures. The proposed model is evaluated on the public BreakHis dataset. Across all four magnification levels, EVC-Net demonstrates robust performance, achieving an average accuracy of 0.985 and an AUC-ROC of 0.994 for binary (benign vs. malignant) classification. For the eight-subtype multi-class task, the model maintains high efficacy, attaining an average accuracy of 0.954 and an AUC-ROC of 0.980. Furthermore, interpretability analysis using Grad-CAM is conducted, generating heatmaps overlay visualization to understand the rational behind model's predictions. These results demonstrate the potential of the EVC-Net framework to enhance diagnostic accuracy and consistency, offering valuable support for clinical workflows in oncology.

CCS Concepts: • **Computing methodologies** → **Artificial intelligence**; **Computer vision**; **Computer vision tasks**;

Additional Key Words and Phrases: EfficientNetV2S, Vision Transformer, Capsule Networks, Histopathology Image Analysis, Breast Cancer Classification, BreakHis Dataset

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1 Introduction

Breast cancer represents a considerable global health concern. As reported by the World Health Organization, it is a prominent cause of cancer-related mortality globally and the most commonly diagnosed cancer in women [47]. Early and accurate diagnosis plays a crucial role in improving patient survival rates and treatment outcomes. Various imaging modalities, such as mammography, ultrasound, and magnetic resonance imaging, are employed for detecting breast abnormalities.

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Histopathological analysis of tissue biopsies, however, is generally considered the definitive standard for diagnosis [45], providing key insights into cellular morphology and tumor characteristics.

The diagnostic process for breast cancer, despite advancements in medical imaging and pathology [8, 40], faces certain challenges. Manual interpretation of histopathological slides, a common practice, can be labor-intensive and is subject to inter-observer variability, which may be influenced by pathologist experience [26, 48]. This subjectivity can lead to diagnostic inconsistencies and potential delays, particularly in cases of late-stage diagnosis. These factors underscore the need for robust and objective methods to support pathologists and enhance current diagnostic workflows.

Artificial intelligence, particularly **Deep Learning (DL)**, has demonstrated significant potential in the domain of medical image analysis [33]. **Computer-Aided Diagnosis (CAD)** systems that incorporate DL algorithms, notably **Convolutional Neural Networks (CNNs)**, have achieved success in various medical image classification tasks [32, 34]. This success is often attributed to their inherent ability to learn hierarchical features directly from image data. Consequently, extensive research has explored the application of pretrained CNN architectures [6] (e.g., MobileNet, ResNet50V2, EfficientNetB1) for classifying breast histopathological images, aiming to accurately distinguish between benign and malignant conditions. However, the local receptive fields of CNNs limit their ability to capture long-range dependencies. This led to the adoption of **Vision Transformers (ViTs)** [21], which use self-attention to model global context effectively . To harness the complementary strengths of both paradigms, hybrid approaches combining CNNs (for local features) and transformers (for global context) have been proposed. While these hybrid strategies have consistently shown improved performance, they do not explicitly preserve the hierarchical relationships within tissue structures, a key diagnostic element in pathology where the spatial arrangement of cells is critical.

In light of these considerations, this article introduces EVC-Net, a hybrid DL framework for the diagnosis of breast cancer from histopathological images. The proposed EVC-Net system synergistically combines EfficientNetV2S [43], for extracting fine-grained textural features; ViTs [21], for capturing long-range dependencies and global contextual information; and capsule networks [14], for preserving spatial hierarchies. This work evaluates the performance of the EVC-Net framework on the publicly available BreakHis dataset [41], addressing both binary (benign vs. malignant) and multi-class (subtype) classification tasks. The main contributions of this work include:

- The development of a novel hybrid DL architecture, EVC-Net, combining EfficientNetV2S, ViTs, and capsule networks for breast cancer histopathology image classification.
- Comprehensive evaluation of the proposed model on a publicly available dataset (BreakHis) for both binary and multi-class classification, demonstrating its effectiveness.
- Achieving high accuracy, precision, recall, and F1-scores, indicating the potential of the hybrid approach to support diagnostic workflows in oncology.

2 Related Work

The automated analysis of histopathological images for breast cancer diagnosis has evolved significantly, transitioning from traditional machine learning methods to advanced DL architectures [19, 23]. Early approaches required manual extraction of handcrafted features from images such as texture, color, and morphology, which were fed into classifiers like support vector machines, K-nearest neighbors, and random forests to perform classification [22]. For instance, Dass et al. [12] developed a model using K-means clustering and a probabilistic neural network for classification based on a new set of features. While foundational, these methods were often laborious and struggled to capture the full complexity of histological patterns [12, 24]. The advent of DL, particularly CNNs, marked a paradigm shift by enabling automatic and hierarchical feature learning

directly from pixel data [5, 24]. Architectures such as VGG16 [1], ResNet [10], DenseNet [46], and EfficientNet [38] have been extensively applied to breast cancer histopathology [2, 29], setting strong performance benchmarks on public datasets like BreakHis [30, 31, 41]. These models excel at extracting rich, localized features like cellular morphology and texture, which are critical for distinguishing between benign and malignant tissues.

Recent work continues to refine CNN-based approaches. For instance, some studies propose novel CNN blocks, such as the Small-SE-ResNeXt block in the SSResNeXt architecture [49], which integrates asymmetric convolutions and channel attention to improve feature extraction for multi-class classification on the BreakHis dataset without a significant increase in model complexity. Other recent efforts, such as Patho-Net [27], combine established architectures like U-Net with gated recurrent units to handle varying image sizes more effectively and integrate automated color normalization techniques to improve model scalability and robustness. Given the often-limited size of medical imaging datasets, transfer learning has become a standard and effective strategy. This approach involves fine-tuning a model pretrained on a large-scale natural image dataset, such as ImageNet [24], on the specific medical imaging task. This leverages the rich feature representations learned from the large dataset, often leading to improved performance and faster convergence. Han et al. [13] used transfer learning in their CSDCNN model for breast cancer classification, while Khan et al. [20] and Yari et al. [50] also demonstrated the efficacy of pretrained ResNet and DenseNet models.

More recently, research has begun to explore architectures beyond standard CNNs. ViTs, which have excelled in natural language processing, are being adapted for computer vision tasks. ViTs use a self-attention mechanism to capture long-range dependencies and global relationships within an image, a potential advantage over the inherently local receptive fields of CNNs. Models like the ViT [9] and the hierarchical swin transformer [25] have demonstrated competitive performance on medical imaging tasks, including breast cancer classification. For instance, recent studies such as Hossain et al. [16] have specifically explored the use of ViTs for categorizing breast cancer from H&E histopathology images. Their ability to model the global layout of tissue structures is a distinct advantage. Their ability to model the global layout of tissue structures is a distinct advantage. Nonetheless, as observed in this study, while these architectures excel at binary classification, their performance can be less stable when transitioning to more complex, fine-grained multi-class settings, highlighting a potential limitation for standalone transformer models.

To harness the complementary strengths of both paradigms, hybrid approaches combining CNNs and transformers have been proposed. These models typically use a CNN backbone to extract low-level local features, while transformer modules to model global context. This fusion aims to create a more comprehensive feature representation that benefits from both detailed local information and broader contextual understanding. Several recent studies have explored this direction; for example, BreastNet integrates attention modules into a CNN framework [3], and FCCS-Net employs a multi-level attention mechanism [28]. Other innovative methods have explored graph convolutional networks [11] and modified supervised contrastive learning [36] to further improve classification. While these hybrid strategies have consistently shown improved performance, they do not explicitly preserve the hierarchical relationships within tissue structures, a key diagnostic element in pathology where the spatial arrangement of cells is critical [7].

Building on these advancements, our work introduces EVC-Net to address these specific gaps. By synergistically integrating three distinct concepts—a state-of-the-art CNN (EfficientNetV2S [43]) for local feature extraction, a ViT [9] for global context, and a capsule network [35] to explicitly model spatial hierarchies—our framework is designed to create a more complete and discriminative feature representation. This three-pronged architecture aims to overcome the individual limitations of its components, providing a robust and effective approach for histopathological image analysis in both binary and complex multi-class tasks.

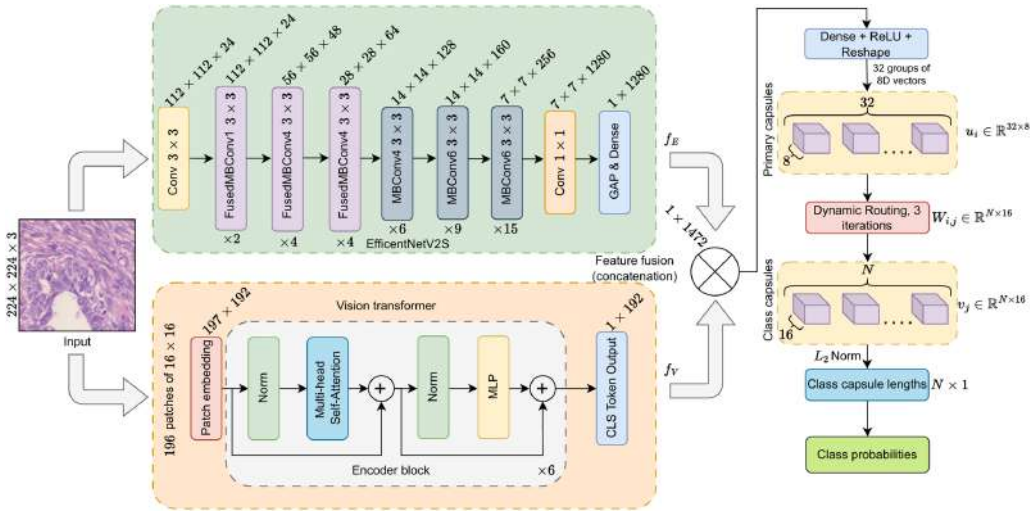


Fig. 1. Overall architecture of the EVC-Net model. The framework comprises parallel EfficientNetV2S and ViT branches for feature extraction. EfficientNetV2S provides local features (f_E), while the ViT (6 encoder blocks) yields global contextual features (f_V). These are fused and processed by a capsule network, involving primary capsules, dynamic routing, and N class capsules, to produce the final classification probabilities for N classes.

3 Proposed Methodology

This section details the proposed hybrid DL framework, EVC-Net designed for the accurate classification of breast cancer from histopathological images. An overview of the EVC-Net architecture is provided in Figure 1. EVC-Net integrates the distinct strengths of EfficientNetV2S for local feature extraction, ViTs for global context comprehension, and capsule networks for modeling hierarchical spatial relationships within tissue structures. Input histopathological images are pre-processed and passed through parallel EfficientNetV2S and ViT branches, followed by feature fusion and further refinement by a capsule network layer before final classification.

3.1 Preprocessing

The experiments are conducted using the publicly available BreakHis dataset [41]. Prior to model input, images are resized to a uniform dimension of 224 × 224 pixels. To augment the training data and improve model generalization, several techniques are applied, including random rotations up to 90 degrees, zooming up to a factor of 1.2, width and height shifts of up to 50%, and both horizontal and vertical flips. These augmentations aim to simulate real-world variations in histopathological slides.

3.2 EfficientNetV2S Branch

The EfficientNetV2 architecture's design rationale emphasizes an improved balance between computational efficiency (training speed, throughput) and parameter efficiency relative to earlier models [42]. This enhancement is achieved partly through incorporating a mix of Fused-MBConv and standard MBConv blocks (Figure 2). This favorable performance-to-compute ratio established EfficientNetV2S as the optimal a priori backbone choice for the proposed hybrid framework, as minimizing the computational demands of individual components is crucial for managing the overall model complexity. Furthermore, this selection provides a high-performance feature representation, the efficacy of which is confirmed by the baseline results presented in the ablation study.

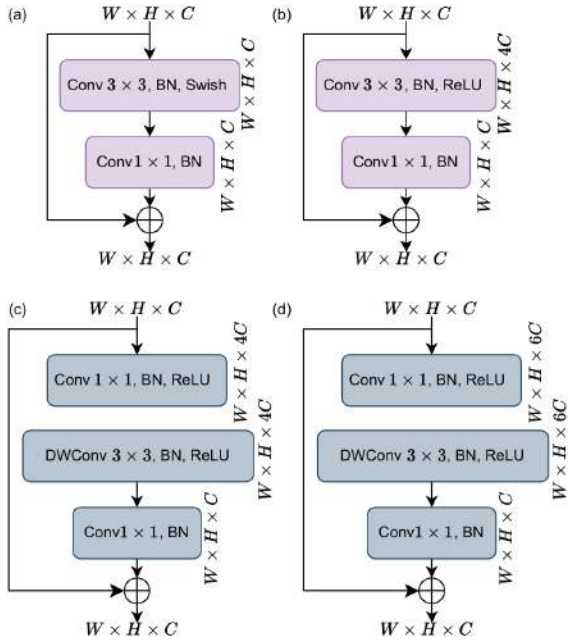


Fig. 2. Schematic representation of components of EfficientNetV2S: (a) Fused-MBConv1 3 × 3, (b) Fused-MBConv4 3 × 3, (c) MBConv4 3 × 3, and (d) MBConv6 3 × 3.

The EfficientNetV2 branch extracts the fine-grained local texture features from the input histopathological images. Let the input image be represented as $X \in \mathbb{R}^{H \times W \times C}$, with dimensions $224 \times 224 \times 3$. The EfficientNetV2S backbone, denoted by the function Φ_E, processes this input through a sequence of Fused-MBConv and MBConv blocks (Figure 2) to generate a high-level feature map. This feature map is subsequently passed through a **Global Average Pooling (GAP)** layer and a fully connected (dense) layer to produce the final local feature vector, f_E (as shown in Equation (1)). The resulting vector, $f_E \in \mathbb{R}^{1,280}$, serves as a compact representation of the local morphological characteristics of the tissue:

$$f_E = \text{Dense}(\text{GAP}(\Phi_E(X))). \tag{1}$$

3.3 ViT Branch

To capture long-range dependencies and the global context of the image, a parallel ViT branch is utilized. The input image X is first partitioned into a sequence of N non-overlapping patches $\{x_p^i\}_{i=1}^P$, where each patch has a resolution of 16×16 pixels, resulting in $P = 196$ patches. These patches are flattened and linearly projected into embeddings. A learnable classification token embedding, x_{class}, is prepended to the sequence of patch embeddings. Positional embeddings, E_{pos}, are added to retain spatial information. The resulting sequence of vectors, z_0, is given by Equation (2):

$$z_0 = [x_{class}; E(x_p^1); E(x_p^2); \ldots; E(x_p^P)] + E_{pos}. \tag{2}$$

Here, E is the learned linear projection. This sequence is then processed by a series of $L = 6$ transformer encoder blocks. The operation within each block l, shown in Equations (3) and (4), consists of **Layer Normalization (LN)**, **Multi-Head Self-Attention (MSA)**, and a **Multilayer Perceptron (MLP)**. These transformer encoder operations follow the standard ViT architecture [9]. The output of the classification token from the final encoder block, z_L^0, is taken as the global

feature representation $f_V \in \mathbb{R}^{192}$:

$$z'_l = \text{MSA}(\text{LN}(z_{l-1})) + z_{l-1}. \tag{3}$$

$$z_l = \text{MLP}(\text{LN}(z'_l)) + z'_l. \tag{4}$$

3.4 Capsule Network Classifier

The feature vectors from the two branches are fused via concatenation, as shown in Equation (5), to create a comprehensive representation that combines both local and global information. Alternative strategies, such as element-wise addition, would be impractical, as they would require an arbitrary projection layer to equalize the vector lengths. The $F_{fused} \in \mathbb{R}^{1,472}$ vector encapsulates both fine-grained local features (f_E) and global contextual features (f_V):

$$F_{fused} = \text{Concat}(f_E, f_V). \tag{5}$$

However, using a standard fully connected classifier at this stage would flatten these features, potentially disregarding the critical spatial relationships between them. In histopathology, the spatial arrangement of cells and tissue components is a key diagnostic feature. Therefore, we employ a capsule network as the classification head specifically to preserve and model these hierarchical spatial relationships within the tissue structures. This fused vector, $F_{fused} \in \mathbb{R}^{1,472}$, is the input to the capsule network. The network first transforms the fused vector into a set of 32 primary capsules $\{u_i\}_{i=1}^{32}$, where each capsule is an 8-dimensional vector. These primary capsules are then routed to a layer of N class capsules (here N =$ N =]]> 2 and N =$ N =]]> 8 for the binary and multi-class classification, respectively), each a 16-dimensional vector, using a dynamic routing algorithm with 3 iterations. For each class capsule j, its input, s_j, is a weighted sum of prediction vectors from all primary capsules, as described in Equation (6):

$$s_j = \sum_i c_{ij} W_{ij} u_i. \tag{6}$$

Here, the coupling coefficients c_{ij} are refined iteratively based on the agreement between the primary capsule outputs and the class capsule outputs. The output vector of each class capsule, v_j, is obtained by applying a non-linear squashing function, shown in Equation (7), which scales the vector length to a value between 0 and 1. This dynamic routing process and squashing function are adopted from the foundational work on capsule networks [35]. The length of each class capsule vector, $\|v_j\|$, represents the probability of the input belonging to class j:

$$v_j = \frac{||s_j||}{1 + ||s_j||^2} s_j. \tag{7}$$

4 Experiment Setup

4.1 Dataset Description

The current study utilizes the Breast Cancer Histopathological Image Classification (BreakHis) dataset, which was released by the Laboratory of Vision, Robotics, and Imaging at the Federal University of Paraná, Brazil [41]. This publicly available dataset is organized into two primary classes: benign and malignant tumors. The histopathological samples were imaged under four distinct optical magnifications (mag.: 1,240×, 100×, 200×, and 400×). The images are provided in RGB format with three color channels, at a resolution of 700 × 460 pixels.

The BreakHis dataset comprises a total of 7,909 images, sourced from 82 patients. These include 2,480 benign samples and 5,429 malignant samples. The benign category is further divided into four subtypes: Adenosis (A), Fibroadenoma (F), Phyllodes Tumor (PT), and Tubular Adenoma (TA). Similarly, the malignant category consists of four subtypes: Ductal Carcinoma (DC), Lobular

Table 1. Detailed Distribution of Images by Subtype and Magnification in the BreakHis Dataset

Mag.	Benign Subtypes				Malignant Subtypes				Total
	A	F	TA	PT	DC	LC	MC	PC	
40×	114	253	109	149	864	156	205	145	1,995
100×	113	260	121	150	903	170	222	142	2,081
200×	111	264	108	140	896	163	196	135	2,013
400×	106	237	115	130	788	137	169	138	1,820
Total	444	1,014	453	569	3,451	626	792	560	7,909
Patients	4	10	3	7	38	5	9	6	82

Carcinoma (LC), Mucinous Carcinoma (MC), and Papillary Carcinoma (PC). A detailed summary of the image distribution across different magnifications is presented in Table 1.

The BreakHis dataset [41] features histopathological images obtained directly from tissue biopsy samples. This characteristic simplifies certain aspects of the classification pipeline, as the images often represent regions of interest, allowing models to focus on learning relevant features from the pixel data without extensive initial segmentation for patch extraction, unlike some other medical imaging modalities such as mammography where tumors may occupy a small portion of the overall image. Histopathological images generally provide a direct and detailed view of tissue morphology. The availability of images at multiple magnification levels offers a rich source of information, presenting fine morphological and structural details of the tissue samples, which is valuable for training DL algorithms. As illustrated in Figure 3, the dataset's inclusion of various benign and malignant breast cancer subtypes across these magnifications provides a comprehensive basis for developing and evaluating DL-based diagnostic models.

4.2 Implementation Details

The proposed EVC-Net model is implemented using Python with PyTorch library. The source code is publicly available at Github[1] repository. The models are trained and evaluated on system equipped with the NVIDIA GeForce RTX 4090 GPU. The EVC-Net model comprises approximately 21.5 million trainable parameters, where the EfficientNetV2S and ViT backbones are initialized with weights pretrained on the ImageNet dataset. All newly added layers, including the feature fusion, capsule network, and dense classification head, were initialized using the He normal method. Key hyperparameters, including the initial learning rate (tested values: 1×10^{-3}, 1×10^{-4}, 1×10^{-5}), batch size (tested values: 64, 128, 256), and dropout rate (tested values: 0.3, 0.4, 0.5), were determined prior to the final 5-fold cross-validation experiments. A grid search methodology is employed using a dedicated validation set (a separate 10% split from the 80% training data partition, not used in the final K-fold testing). The combination yielding the best validation performance (Learning Rate: 1×10^{-4}, Batch Size: 128, Dropout: 0.4) is then used for all subsequent training and evaluation reported in this study. Training for each classification task (binary and multi-class) was conducted for a maximum of 120 epochs. To dynamically adjust the learning rate during training, the "ReduceLROnPlateau" callback is employed; this callback monitored the validation loss and reduced the learning rate when a plateau is detected, facilitating more effective convergence.

For the binary classification task (benign vs. malignant), the binary cross-entropy loss function is utilized (shown in Equation (8)). For the multi-class classification task (differentiating between

[1]https://github.com/nspunn1993/EVC-Net/.

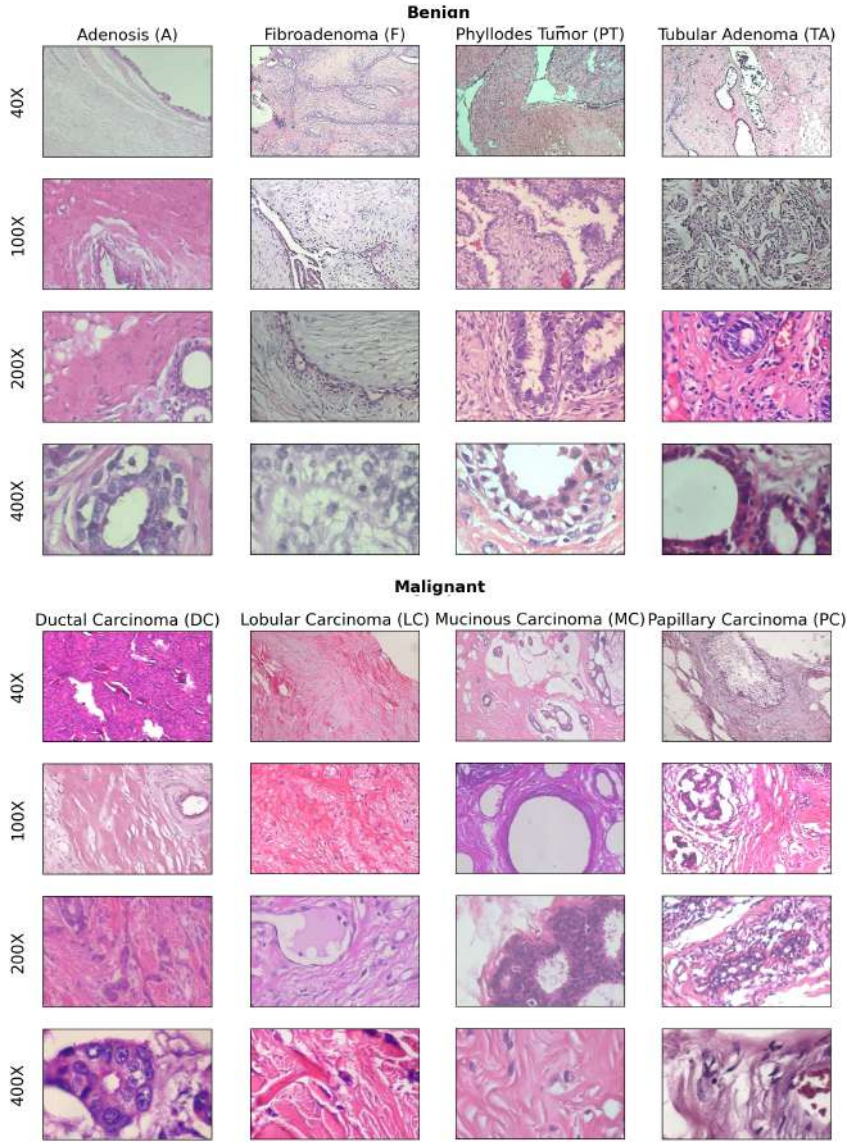


Fig. 3. Sample of distinct types of breast cancer histopathology images from the BreakHis dataset.

eight breast cancer subtypes), the categorical cross-entropy loss function (shown in Equation (9)) is applied. Regularization techniques are incorporated to mitigate overfitting. An early stopping mechanism is implemented, to monitor the validation loss and terminate training if no improvement is observed for 10 number of consecutive epochs, thus retaining the model weights from the epoch with the best validation performance. Furthermore, dropout layers with a dropout rate of 0.4 are applied after the dense layers in the classification head to enhance model generalization:

$$L_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)], \tag{8}$$

$$L_{CCE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K} y_{ik}\log(p_{ik}), \tag{9}$$

where y_i is the true label (0 or 1) for sample i, p_i is the predicted probability for class 1 for sample i, y_{ik} is an indicator function (1 if sample i belongs to class k, and 0 otherwise), and p_{ik} is the predicted probability that sample i belongs to class k. Both loss functions are standard formulations bounded below by 0, heavily penalizing high-confidence misclassifications.

To ensure a robust evaluation and reliable assessment, a 5-fold cross-validation is performed. The dataset, which comprises images from 82 patients, is partitioned at the patient level. This ensures that all images from a single patient are assigned to the same fold, preventing data leakage between the training and testing sets. Within each fold, 80% of the patient data is used for training and 20% for validation. The final performance metrics are reported as the mean and standard deviation across all five folds. To optimize computational speed and reduce memory footprint without significantly compromising model accuracy, mixed-precision training is enabled. Model checkpoints are saved periodically during training, specifically when an improvement in validation performance is achieved. Performance is benchmarked for both binary and multi-class classification tasks with standard evaluation metrics: accuracy (Acc.), precision (Pre.), recall (Rec.), F1-score (F1), and the area under the receiver operating characteristic curve (AUC-ROC or AR).

5 Results and Discussion

The proposed EVC-Net framework is evaluated on the BreakHis dataset for both binary and multi-class classification tasks. The comprehensive performance of the proposed model against several baseline CNN-based models (e.g., EfficientNetB1, DenseNet201, ResNet variants) and transformer-based architectures (ViT, swin transformer) across all four magnifications (40×, 100×, 200×, and 400×) is presented in Tables 2 and 3. To systematically validate our architecture, a detailed ablation study is presented in Table 4, quantifying the contribution of each EVC-Net component. Additionally, for a direct and fair benchmark, a comparison against several recent methods with publicly available code is detailed in Table 5. Finally, a fine-grained, class-wise performance analysis for the eight-subtype multi-class task is provided in Table 6.

5.1 Comparison with State-of-the-Art Models

In binary classification, EVC-Net achieves consistently high performance across all evaluated magnifications. As indicated in Table 2, EVC-Net mean accuracies range from 0.983 (at 100×) to a peak of 0.988 (at 200×), with corresponding mean AUC-ROC values consistently above 0.990 and reaching up to 0.997 (at 200×). This level of performance is generally higher than that of the benchmark models listed in Table 2 for each respective magnification group. For instance, while ViT-Base also records a mean accuracy of 0.988 at 200×, EVC-Net maintains consistently high mean AUC-ROC scores across all magnifications, indicating strong discrimination capabilities irrespective of the specific magnification factor.

For multi-class classification, EVC-Net again demonstrates consistent high performance across the magnifications. Mean accuracies range from 0.950 (at 400×) to 0.958 (at 40×), with corresponding mean AUC-ROC values between 0.976 (at 400×) and 0.984 (at 40×), as detailed in Table 3. When comparing the transition from binary to the more challenging multi-class setting, EVC-Net shows higher stability. Calculating the average mean accuracy across the four magnifications, EVC-Net's accuracy drops by approximately 3.20% from binary (average 0.9853) to multi-class (average 0.9538) tasks. In contrast, ViT-Base exhibits a more pronounced decrease in average mean accuracy of approximately 6.93% (from an average of 0.9850 in binary to 0.9168 in multi-class tasks

Table 2. Performance for Binary (Benign vs. malignant) Classification on the BreakHis Test Set

Method	Mag.	Acc.	Pre.	Rec.	F1	AR
MobileNetV3 [17]	40×	0.980 ± 0.004	0.981 ± 0.004	0.977 ± 0.005	0.979 ± 0.004	0.989 ± 0.002
	100×	0.979 ± 0.005	0.980 ± 0.004	0.976 ± 0.005	0.978 ± 0.005	0.988 ± 0.003
	200×	0.985 ± 0.003	0.986 ± 0.003	0.982 ± 0.004	0.984 ± 0.003	0.993 ± 0.002
	400×	0.983 ± 0.004	0.984 ± 0.003	0.980 ± 0.005	0.982 ± 0.004	0.991 ± 0.002
EfficientNetB1 [42]	40×	**0.984 ± 0.003**	0.984 ± 0.003	**0.981 ± 0.004**	0.982 ± 0.003	**0.992 ± 0.002**
	100×	0.982 ± 0.004	0.983 ± 0.004	**0.980 ± 0.005**	0.981 ± 0.004	0.990 ± 0.003
	200×	**0.988 ± 0.002**	0.988 ± 0.002	0.985 ± 0.003	0.986 ± 0.002	0.995 ± 0.001
	400×	0.985 ± 0.003	0.986 ± 0.003	**0.983 ± 0.004**	0.984 ± 0.003	0.993 ± 0.002
VGG16 [39]	40×	0.973 ± 0.006	0.977 ± 0.005	0.967 ± 0.007	0.972 ± 0.006	0.987 ± 0.004
	100×	0.968 ± 0.007	0.972 ± 0.006	0.962 ± 0.008	0.967 ± 0.007	0.982 ± 0.005
	200×	0.970 ± 0.006	0.974 ± 0.006	0.964 ± 0.007	0.969 ± 0.006	0.984 ± 0.004
	400×	0.969 ± 0.007	0.973 ± 0.006	0.963 ± 0.008	0.968 ± 0.007	0.983 ± 0.005
ResNet50V2 [15]	40×	0.978 ± 0.005	0.980 ± 0.004	0.974 ± 0.006	0.977 ± 0.005	0.988 ± 0.003
	100×	0.977 ± 0.005	0.979 ± 0.005	0.973 ± 0.006	0.976 ± 0.005	0.987 ± 0.003
	200×	0.983 ± 0.004	0.985 ± 0.003	0.979 ± 0.005	0.982 ± 0.004	0.992 ± 0.002
	400×	0.980 ± 0.005	0.982 ± 0.004	0.976 ± 0.006	0.979 ± 0.005	0.990 ± 0.003
ResNet152V2 [15]	40×	0.980 ± 0.004	0.982 ± 0.004	0.976 ± 0.005	0.979 ± 0.004	0.989 ± 0.003
	100×	0.979 ± 0.005	0.981 ± 0.004	0.975 ± 0.006	0.978 ± 0.005	0.988 ± 0.003
	200×	0.985 ± 0.003	0.987 ± 0.003	0.981 ± 0.004	0.984 ± 0.003	0.993 ± 0.002
	400×	0.982 ± 0.004	0.984 ± 0.004	0.978 ± 0.005	0.981 ± 0.004	0.991 ± 0.002
DenseNet201 [18]	40×	0.983 ± 0.004	0.985 ± 0.003	0.971 ± 0.007	0.978 ± 0.005	0.991 ± 0.002
	100×	**0.983 ± 0.004**	0.984 ± 0.003	0.970 ± 0.007	0.977 ± 0.005	0.990 ± 0.003
	200×	**0.988 ± 0.002**	0.989 ± 0.002	0.976 ± 0.006	0.982 ± 0.004	0.995 ± 0.001
	400×	0.985 ± 0.003	0.987 ± 0.003	0.982 ± 0.004	0.984 ± 0.003	0.993 ± 0.002
ViT-Base [9]	40×	**0.984 ± 0.004**	0.978 ± 0.005	0.977 ± 0.006	0.977 ± 0.005	0.989 ± 0.003
	100×	**0.983 ± 0.004**	0.977 ± 0.005	0.976 ± 0.006	0.976 ± 0.005	0.988 ± 0.003
	200×	**0.988 ± 0.003**	0.983 ± 0.004	0.982 ± 0.004	0.982 ± 0.004	0.993 ± 0.002
	400×	0.985 ± 0.003	0.980 ± 0.005	0.979 ± 0.005	0.979 ± 0.005	0.991 ± 0.002
Swin transformer [25]	40×	0.979 ± 0.005	0.981 ± 0.004	0.976 ± 0.006	0.978 ± 0.005	0.988 ± 0.003
	100×	0.978 ± 0.005	0.980 ± 0.005	0.975 ± 0.006	0.977 ± 0.005	0.987 ± 0.004
	200×	0.984 ± 0.004	0.986 ± 0.003	0.981 ± 0.005	0.983 ± 0.004	0.992 ± 0.002
	400×	0.981 ± 0.005	0.983 ± 0.004	0.978 ± 0.006	0.980 ± 0.005	0.990 ± 0.003
EVC-Net (ours)	40×	**0.984 ± 0.003**	**0.986 ± 0.003**	**0.981 ± 0.004**	**0.983 ± 0.003**	**0.992 ± 0.002**
	100×	**0.983 ± 0.004**	**0.985 ± 0.003**	**0.980 ± 0.005**	**0.982 ± 0.004**	**0.991 ± 0.002**
	200×	**0.988 ± 0.002**	**0.990 ± 0.002**	**0.986 ± 0.003**	**0.988 ± 0.002**	**0.997 ± 0.001**
	400×	**0.986 ± 0.003**	**0.988 ± 0.002**	**0.983 ± 0.004**	**0.985 ± 0.003**	**0.995 ± 0.001**

Values are reported as mean ± standard deviation from 5-fold patient-level cross-validation. Bold quantities indicate the maximum mean value for each metric within each magnification (Mag.) group.

across magnifications). This indicates that while established architectures perform well in binary tasks, EVC-Net maintains higher efficacy in the fine-grained multi-class setting across different magnifications. These results underscore the contribution of EVC-Net's convolutional components in preserving local feature extraction, which is crucial for distinguishing between multiple similar subtypes under varying visual scales.

Training dynamics further illustrate the model's behavior. Figure 4 displays the training and validation cross-entropy loss curves for the 40× magnification experiments for both binary and multi-class tasks. Both curves demonstrate effective learning, with the training loss consistently decreasing and the validation loss converging to a stable minimum. The early stopping mechanism

Table 3. Performance for Multi-Class (Eight Subtypes) Classification on the BreakHis Test Set

Method	Mag.	Acc.	Pre.	Rec.	F1	AR
MobileNetV3 [17]	40×	0.928 ± 0.009	0.931 ± 0.008	0.923 ± 0.010	0.927 ± 0.009	0.968 ± 0.005
	100×	0.922 ± 0.011	0.925 ± 0.010	0.917 ± 0.011	0.921 ± 0.010	0.962 ± 0.007
	200×	0.925 ± 0.010	0.928 ± 0.009	0.920 ± 0.010	0.924 ± 0.009	0.965 ± 0.006
	400×	0.920 ± 0.011	0.923 ± 0.011	0.915 ± 0.012	0.919 ± 0.011	0.960 ± 0.007
EfficientNetB1 [42]	40×	0.938 ± 0.008	0.945 ± 0.007	0.933 ± 0.009	0.939 ± 0.008	0.973 ± 0.004
	100×	0.932 ± 0.009	0.939 ± 0.008	0.927 ± 0.010	0.933 ± 0.009	0.967 ± 0.006
	200×	0.935 ± 0.009	0.942 ± 0.008	0.930 ± 0.010	0.936 ± 0.009	0.970 ± 0.005
	400×	0.930 ± 0.010	0.937 ± 0.009	0.925 ± 0.011	0.931 ± 0.010	0.965 ± 0.006
VGG16 [39]	40×	0.893 ± 0.014	0.898 ± 0.013	0.883 ± 0.016	0.890 ± 0.015	0.943 ± 0.009
	100×	0.887 ± 0.016	0.892 ± 0.015	0.877 ± 0.017	0.884 ± 0.016	0.937 ± 0.010
	200×	0.890 ± 0.015	0.895 ± 0.014	0.880 ± 0.017	0.887 ± 0.016	0.940 ± 0.010
	400×	0.885 ± 0.017	0.890 ± 0.016	0.875 ± 0.018	0.882 ± 0.017	0.935 ± 0.011
ResNet50V2 [15]	40×	0.933 ± 0.009	0.936 ± 0.008	0.927 ± 0.010	0.931 ± 0.009	0.971 ± 0.005
	100×	0.927 ± 0.010	0.930 ± 0.009	0.921 ± 0.011	0.925 ± 0.010	0.965 ± 0.006
	200×	0.930 ± 0.010	0.933 ± 0.009	0.924 ± 0.011	0.928 ± 0.010	0.968 ± 0.005
	400×	0.925 ± 0.011	0.928 ± 0.010	0.919 ± 0.012	0.923 ± 0.011	0.963 ± 0.007
ResNet152V2 [15]	40×	0.935 ± 0.009	0.938 ± 0.008	0.929 ± 0.010	0.933 ± 0.009	0.973 ± 0.004
	100×	0.929 ± 0.010	0.932 ± 0.009	0.923 ± 0.011	0.927 ± 0.010	0.967 ± 0.006
	200×	0.932 ± 0.009	0.935 ± 0.008	0.926 ± 0.010	0.930 ± 0.009	0.970 ± 0.005
	400×	0.927 ± 0.010	0.930 ± 0.009	0.921 ± 0.011	0.925 ± 0.010	0.965 ± 0.006
DenseNet201 [18]	40×	0.944 ± 0.007	0.941 ± 0.008	0.932 ± 0.010	0.936 ± 0.009	0.978 ± 0.003
	100×	0.938 ± 0.008	0.935 ± 0.009	0.926 ± 0.011	0.930 ± 0.010	0.972 ± 0.005
	200×	0.941 ± 0.008	0.938 ± 0.008	0.929 ± 0.010	0.933 ± 0.009	0.975 ± 0.004
	400×	0.936 ± 0.009	0.933 ± 0.009	0.924 ± 0.011	0.928 ± 0.010	0.970 ± 0.005
ViT-Base [9]	40×	0.921 ± 0.011	0.923 ± 0.010	0.918 ± 0.012	0.920 ± 0.011	0.963 ± 0.007
	100×	0.915 ± 0.012	0.917 ± 0.011	0.912 ± 0.013	0.914 ± 0.012	0.957 ± 0.008
	200×	0.918 ± 0.012	0.920 ± 0.011	0.915 ± 0.013	0.917 ± 0.012	0.960 ± 0.007
	400×	0.913 ± 0.013	0.915 ± 0.012	0.910 ± 0.014	0.912 ± 0.013	0.955 ± 0.008
Swin transformer [25]	40×	0.924 ± 0.010	0.927 ± 0.009	**0.943 ± 0.008**	0.934 ± 0.009	0.965 ± 0.006
	100×	0.918 ± 0.012	0.921 ± 0.011	0.936 ± 0.009	0.928 ± 0.010	0.959 ± 0.008
	200×	0.921 ± 0.011	0.924 ± 0.010	0.939 ± 0.009	0.931 ± 0.010	0.962 ± 0.007
	400×	0.916 ± 0.012	0.919 ± 0.011	0.934 ± 0.010	0.926 ± 0.011	0.957 ± 0.008
EVC-Net (ours)	40×	**0.958 ± 0.005**	**0.948 ± 0.006**	**0.943 ± 0.007**	**0.945 ± 0.006**	**0.984 ± 0.003**
	100×	**0.952 ± 0.006**	**0.942 ± 0.007**	**0.937 ± 0.008**	**0.939 ± 0.007**	**0.978 ± 0.004**
	200×	**0.955 ± 0.005**	**0.945 ± 0.006**	**0.940 ± 0.007**	**0.942 ± 0.006**	**0.981 ± 0.003**
	400×	**0.950 ± 0.006**	**0.940 ± 0.007**	**0.935 ± 0.008**	**0.937 ± 0.007**	**0.976 ± 0.004**

Values are reported as mean ± standard deviation from 5-fold patient-level cross-validation. Bold quantities indicate the maximum mean value for each metric within each magnification (Mag.) group.

terminated training at epoch 63 for the binary task and epoch 73 for the multi-class task (averaged across folds), corresponding to the points of optimal validation performance and preventing significant overfitting.

Furthermore, while other CNN-based models like EfficientNetB1 and DenseNet201 achieve high performance, and transformers such as swin transformer show high recall in specific instances (e.g., 0.943 for multi-class at 40×), EVC-Net generally maintains a balanced performance profile, often achieving higher mean metrics, particularly in AUC-ROC. For example, EVC-Net's mean AUC-ROC values are consistently high across all magnifications, peaking at 0.997 for binary classification (200×) and 0.984 for multi-class classification (40×). The ROC curves, presented in Figure 5, provide

Table 4. Ablation Study of the Proposed EVC-Net Components (E: EfficientNetV2S, V: vision Transformer, C: capsule Network) on the BreakHis Dataset for Binary and Multi-Class Classification (BC and MC) at 40× Magnification

E	V	C	Task	Acc.	Pre.	Rec.	F1	AR
✓			BC	0.965 ± 0.006	0.968 ± 0.005	0.960 ± 0.007	0.964 ± 0.006	0.980 ± 0.004
			MC	0.920 ± 0.011	0.922 ± 0.010	0.910 ± 0.012	0.916 ± 0.011	0.955 ± 0.007
	✓		BC	0.962 ± 0.007	0.965 ± 0.006	0.958 ± 0.008	0.961 ± 0.007	0.978 ± 0.005
			MC	0.915 ± 0.012	0.918 ± 0.011	0.905 ± 0.014	0.911 ± 0.013	0.950 ± 0.008
✓		✓	BC	0.973 ± 0.005	0.976 ± 0.004	0.968 ± 0.006	0.972 ± 0.005	0.987 ± 0.003
			MC	0.935 ± 0.009	0.933 ± 0.008	0.928 ± 0.010	0.930 ± 0.009	0.970 ± 0.005
	✓	✓	BC	0.969 ± 0.006	0.971 ± 0.005	0.964 ± 0.007	0.967 ± 0.006	0.983 ± 0.004
			MC	0.928 ± 0.010	0.925 ± 0.009	0.922 ± 0.011	0.923 ± 0.010	0.964 ± 0.006
✓	✓		BC	0.979 ± 0.004	0.981 ± 0.004	0.976 ± 0.005	0.978 ± 0.004	0.990 ± 0.002
			MC	0.948 ± 0.007	0.940 ± 0.008	0.935 ± 0.009	0.937 ± 0.008	0.975 ± 0.004
✓	✓	✓	BC	**0.984 ± 0.003**	**0.986 ± 0.003**	**0.981 ± 0.004**	**0.983 ± 0.003**	**0.992 ± 0.002**
			MC	**0.958 ± 0.005**	**0.948 ± 0.006**	**0.943 ± 0.007**	**0.945 ± 0.006**	**0.984 ± 0.003**

Values are reported as mean ± standard deviation from 5-fold patient-level cross-validation. Bold quantities indicate the best results.

Table 5. Quantitative Comparison of EVC-Net with Prevailing Methods for Breast Cancer Classification

Study	Method	Task	Acc.	Prec.	Rec.	F1	AR
Toğaça et al. [44]	BreastNet	BC	0.979 ± 0.005	0.976 ± 0.006	0.976 ± 0.006	0.977 ± 0.005	0.989 ± 0.003
		MC	0.941 ± 0.008	0.943 ± 0.008	0.939 ± 0.009	0.941 ± 0.008	0.978 ± 0.004
Maurya et al. [28]	FCCS-Net	BC	0.973 ± 0.006	0.975 ± 0.006	0.971 ± 0.007	0.973 ± 0.006	0.985 ± 0.004
		MC	0.932 ± 0.010	0.934 ± 0.009	0.931 ± 0.010	0.933 ± 0.009	0.969 ± 0.006
Bera et al. [4]	CS-GCN	BC	0.980 ± 0.004	0.981 ± 0.004	0.979 ± 0.005	0.980 ± 0.004	0.989 ± 0.003
		MC	0.943 ± 0.007	0.944 ± 0.007	0.942 ± 0.008	0.943 ± 0.007	0.975 ± 0.005
Sani et al. [36]	ISCL	BC	0.947 ± 0.009	0.951 ± 0.008	0.942 ± 0.010	0.946 ± 0.009	0.979 ± 0.005
		MC	0.895 ± 0.015	0.898 ± 0.014	0.892 ± 0.016	0.895 ± 0.015	0.952 ± 0.008
This work	EVC-Net	BC	0.984± 0.003	0.986± 0.003	0.981± 0.004	0.983± 0.003	0.992± 0.002
		MC	0.958± 0.005	0.948± 0.006	0.943± 0.007	0.945± 0.006	0.984± 0.003

The evaluation metrics are reported as mean ± standard deviation from 5-fold patient-level cross-validation for both binary (BC) and multi-class (MC) tasks on the BreakHis dataset at 40× magnification.

a visual comparison of the discriminative ability of EVC-Net against baseline models at 40× magnification. For binary classification (Figure 5(a)), the EVC-Net curve closely follows the top-left corner, reflecting its high mean AUC of 0.992 and superior performance. Similarly, in the multi-class scenario (Figure 5(b)), the micro-average ROC curve for EVC-Net demonstrates the highest performance with a mean AUC of 0.984, visually confirming its effectiveness compared to the other architectures.

Moreover, the consistently low standard deviations observed for EVC-Net (as detailed in Tables 2 and 3) highlight the model's stability and reliable performance across the different patient-level folds, indicating it is not overly sensitive to a specific data split. These findings support the effectiveness of EVC-Net's hybrid design. The integration of convolutional local feature extraction (EfficientNetV2S), transformer-based global attention (ViT), and capsule network spatial modeling yields a comprehensive feature representation that is effective across different scales of visual information presented by varying magnifications. The consistently high AUC-ROC performance across these conditions is particularly relevant for medical image analysis, indicating the model's reliability across different

Table 6. Class-Wise Performance of the EVC-Net Model for Multi-Class Classification (8 Subtypes) on the BreakHis Dataset at 40× Magnification

Breast cancer subtype	Pre.	Rec.	F1	AR
Benign subtypes				
Adenosis (A)	0.935 ± 0.008	0.920 ± 0.010	0.927 ± 0.009	0.975 ± 0.005
Fibroadenoma (F)	0.950 ± 0.007	0.945 ± 0.008	0.947 ± 0.007	0.985 ± 0.003
Phyllodes Tumor (PT)	0.920 ± 0.010	0.910 ± 0.011	0.915 ± 0.010	0.970 ± 0.006
Tubular Adenoma (TA)	0.940 ± 0.008	0.930 ± 0.009	0.935 ± 0.008	0.980 ± 0.004
Malignant subtypes				
Ductal Carcinoma (DC)	0.965 ± 0.006	0.970 ± 0.005	0.967 ± 0.005	0.990 ± 0.002
Lobular Carcinoma (LC)	0.915 ± 0.011	0.890 ± 0.015	0.902 ± 0.013	0.965 ± 0.007
Mucinous Carcinoma (MC)	0.955 ± 0.007	0.940 ± 0.009	0.947 ± 0.008	0.986 ± 0.003
Papillary Carcinoma (PC)	0.930 ± 0.009	0.915 ± 0.012	0.922 ± 0.010	0.972 ± 0.006
Macro Average	0.948± 0.006	0.943± 0.007	0.945± 0.006	0.984± 0.003

Values are reported as mean ± standard deviation from 5-fold patient-level cross-validation.

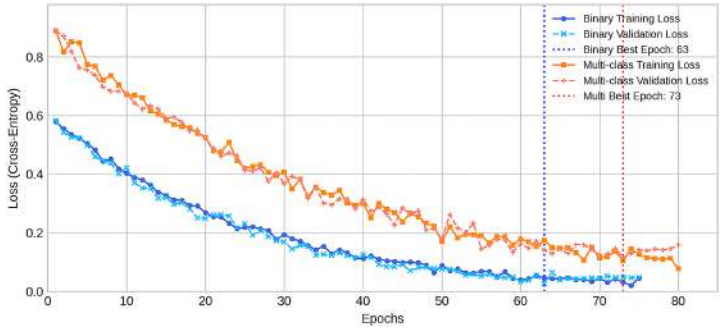


Fig. 4. Training and validation cross-entropy loss curves for the EVC-Net model on binary and multi-class tasks at 40× magnification (averaged over 5 folds). Vertical dotted lines indicate the average epoch selected by early stopping based on the minimum validation loss achieved during training.

decision thresholds and its capability to handle the complex, fine-grained histopathological patterns that benefit from both localized feature detail and global contextual understanding.

5.2 Ablation Study

An ablation study is conducted to systematically evaluate the contribution of each primary component of the EVC-Net: EfficientNetV2S (E), ViT (V), and capsule network (C). These experiments are performed on the BreakHis dataset for both binary (BC) and multi-class (MC) classification tasks, specifically utilizing images at 40× magnification. The detailed performance of various model configurations is presented in Table 4.

First, baseline performance is established using the individual feature extractors. The E configuration (EfficientNetV2S alone) achieves a mean accuracy of 0.965 in binary classification and 0.920 in multi-class classification. The V configuration (ViT alone) yields slightly lower mean accuracies of 0.962 (BC) and 0.915 (MC). Next, the contribution of the capsule network (C) is assessed by adding it as a classifier to each backbone. This integration results in clear performance gains. The E + C configuration improves mean accuracy to 0.973 (BC) and 0.935 (MC). Similarly, the V + C configuration improves mean accuracy to 0.969 (BC) and 0.928 (MC). These improvements over

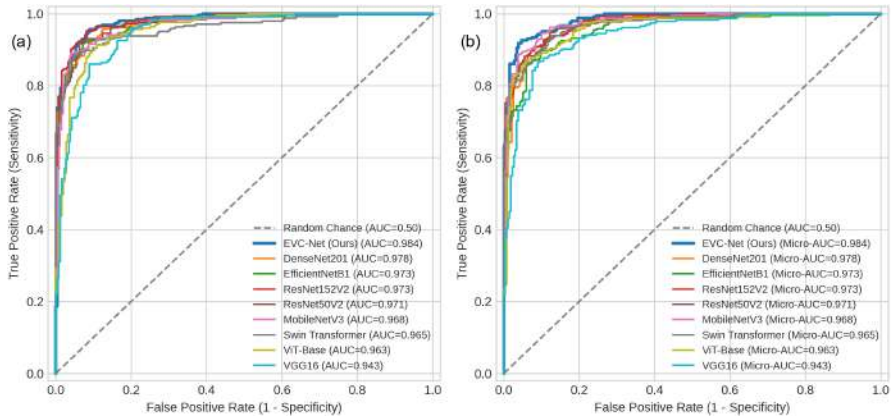


Fig. 5. Receiver Operating Characteristic (ROC) curves comparing EVC-Net against baseline models at 40× magnification. (a) Binary classification performance. (b) Multi-class classification performance using the micro-average method.

the E-only and V-only models indicate that the capsule network's capacity for modeling spatial hierarchies contributes to an enhanced feature representation.

To evaluate the core fusion strategy, the two feature extractors are combined (the E + V configuration) without the capsule classifier. This fusion of local (E) and global (V) features leads to a notable improvement over the single-backbone models, achieving a mean accuracy of 0.979 (BC) and 0.948 (MC). This outcome highlights the significant benefit derived from fusing local feature extraction with global contextual modeling. Finally, the complete EVC-Net (E + V + C) model, which integrates all three components, achieves the highest mean performance metrics. At 40× magnification, it attains a mean accuracy of 0.984 (AUC-ROC 0.992) for binary classification and a mean accuracy of 0.958 (AUC-ROC 0.984) for the multi-class setting. These performance metrics, detailed in Table 4, are superior to those of all other ablated configurations. Furthermore, the complete (E + V + C) model also exhibits the lowest standard deviations (e.g., 0.003 for BC Acc. and 0.005 for MC Acc.). This demonstrates that the synergistic integration of all three components not only yields the highest accuracy but also the most stable and robust framework, confirming the effective contribution of each component.

5.3 Comparison with Prevailing Work

To further validate the performance of the EVC-Net framework, a direct comparison is made against several recent and reproducible methods at 40× magnification. The quantitative results of this benchmark are summarized in Table 5. This analysis includes diverse architectural paradigms such as BreastNet[2] [44], which integrates attention modules; FCCS-Net[3] [28], which uses a multi-level attention approach; CS-GCN[4] [4], a model based on graph convolutional networks; and ISCL[5] [36], which employs a modified supervised contrastive learning method.

In the binary classification task, the proposed EVC-Net achieves a mean accuracy of 98.40% and a mean AUC-ROC of 99.20%. As shown in Table 5, this result exceeds the performance of other methods, including the graph-based CS-GCN [4] (98.00% mean accuracy) and the attention-based

[2]https://github.com/Goodsea/BreastNet.
[3]https://github.com/maurya123ritesh47/FCCS-Net.
[4]https://github.com/asish-bera/CS-GCN.
[5]https://github.com/matinamehdizadeh/Breast-Cancer-Detection.

BreastNet [44] (97.99% mean accuracy). Furthermore, EVC-Net's low standard deviation (e.g., 0.003 for accuracy) signifies high stability across data folds, a desirable characteristic for reproducible clinical-grade tools. This indicates a high level of discriminative capability between benign and malignant classes.

A notable performance differential is observed in the more challenging multi-class scenario. The proposed EVC-Net achieves a mean accuracy of 95.80%, which is higher than the mean accuracies achieved by the next best-performing reproducible models, CS-GCN (94.30%) and BreastNet (94.10%). This outcome reinforces the finding that the EVC-Net's combination of local texture analysis (EfficientNetV2S), global contextual understanding (ViT), and spatial hierarchy modeling (capsule network) provides a more comprehensive and discriminative feature representation.

5.4 Classwise Performance

A detailed analysis of the EVC-Net model's performance across individual classes in the multi-class scenario (eight subtypes) at 40× magnification is presented in Table 6. This class-wise evaluation provides specific insights into the model's efficacy for each breast cancer subtype.

The EVC-Net model demonstrates consistent performance across most subtypes. For the benign classes, "F" yields high metrics, with a mean F1-score of 0.947 and a mean AUC-ROC of 0.985. "A" and "TA" also show robust classification with mean F1-scores of 0.927 and 0.935, and mean AUC-ROC values of 0.975 and 0.980, respectively. "PT" records a mean F1-score of 0.915 and a mean AUC-ROC of 0.970, indicating a slightly lower but still effective classification for this benign subtype. Among the malignant subtypes, "DC" is classified with the highest efficacy, achieving a mean F1-score of 0.967 and a mean AUC-ROC of 0.990. This reflects strong discrimination for this common malignant class. "MC" also exhibits high performance with a mean F1-score of 0.947 and a mean AUC-ROC of 0.986. "PC" obtains a mean F1-score of 0.922 and a mean AUC-ROC of 0.972. "LC" presents with a mean F1-score of 0.902, influenced by a mean recall of 0.890, while maintaining a mean AUC-ROC of 0.965.

Further insight into the model's classification behavior at 40× magnification is provided by the confusion matrices in Figure 6. The binary confusion matrix (Figure 6(a)) shows very high true positive (98.54%) and true negative (98.41%) rates, with minimal misclassifications between benign and malignant classes. The multi-class confusion matrix (Figure 6(b)) highlights the model's strong performance on most classes, particularly DC (97.69% correct) and F (98.04% correct). It also visually confirms areas of slightly higher confusion, such as the misclassification between LC and DC (6.45% of LC predicted as DC) and between TA and A/PT (4.55% each), aligning with the quantitative results discussed previously. This detailed class-wise analysis is valuable for identifying specific areas where model performance is particularly strong or where further targeted improvements, such as class-specific data augmentation or feature engineering, might be beneficial.

The observed variations in performance metrics across the different classes are attributed to the distinct morphological characteristics and inter-class similarities inherent to each subtype, as well as potential variations in the number of training samples per class. For example, the lower mean recall for "LC" might suggest a higher rate of misclassification of "LC" instances as other subtypes, potentially due to less distinct features or fewer representative samples compared to a class like "DC." Nevertheless, the per-class metrics collectively contribute to the overall macro average scores (mean Precision: 0.948, mean Recall: 0.943, mean F1-score: 0.945, mean AUC-ROC: 0.984) better than other models (as shown in Table 3), confirming the EVC-Net model's general capability for fine-grained differentiation among the eight breast cancer subtypes. Moreover, the consistently low standard deviations across most classes, particularly the high-performing "DC" and "F" subtypes, underscore the model's reliability in fine-grained differentiation. This detailed class-wise analysis is valuable for identifying specific areas where model performance is particularly strong or where

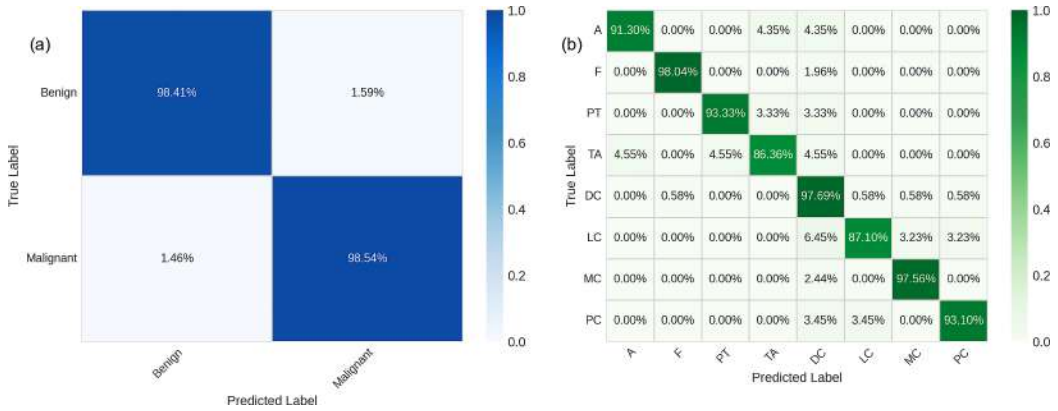


Fig. 6. Normalized confusion matrices for the EVC-Net model at 40× magnification (averaged over 5 folds) test set. Values represent the percentage of the true class (row) predicted as each specific class (column). (Left) Binary classification. (Right) Multi-class classification across the eight subtypes.

further targeted improvements, such as class-specific data augmentation or feature engineering, might be beneficial.

5.5 Interpretability

To gain insights into the decision-making process of the proposed EVC-Net model and to understand which image regions are most influential for its predictions, a common approach involves utilizing class activation mapping techniques such as Grad-CAM [37]. These methods generate heatmaps that highlight the discriminative image regions attended to by the network when classifying an input sample.

Figure 7 presents a selection of histopathological images from the BreakHis dataset alongside their corresponding Grad-CAM heatmaps generated from the EVC-Net model. The heatmaps are overlaid on the input images, where the color intensity correlates with the degree of model attention. Specifically, brighter colors (progressing from yellow to green) highlight regions that the EVC-Net model deems more critical for its classification decision, while darker colors (progressing from blue to purple) indicate less salient regions. For instance, in Figure 7, the heatmaps indicate that EVC-Net focuses on specific cellular arrangements, glandular structures, or stromal regions pertinent to the respective subtype classification. The analysis of these visual explanations is important for qualitatively verifying that the model learns diagnostically relevant features. By examining the highlighted areas, it is possible to assess whether EVC-Net's attention aligns with established pathological indicators, such as nuclear morphology, architectural patterns of cell clusters, or characteristics of the tumor microenvironment, rather than focusing on potential imaging artifacts or irrelevant background areas. This alignment provides confidence in the model's learned representations. Furthermore, such interpretability is valuable for understanding model behavior, particularly in cases of misclassification. Heatmaps can reveal if incorrect predictions stem from the model attending to ambiguous or misleading image features, thereby offering directions for model refinement or identifying challenging diagnostic scenarios. The consistent localization of salient regions by EVC-Net across various subtypes and magnifications, as shown in Figure 7, aids in understanding the model's operational basis and supports its potential utility in CAD.

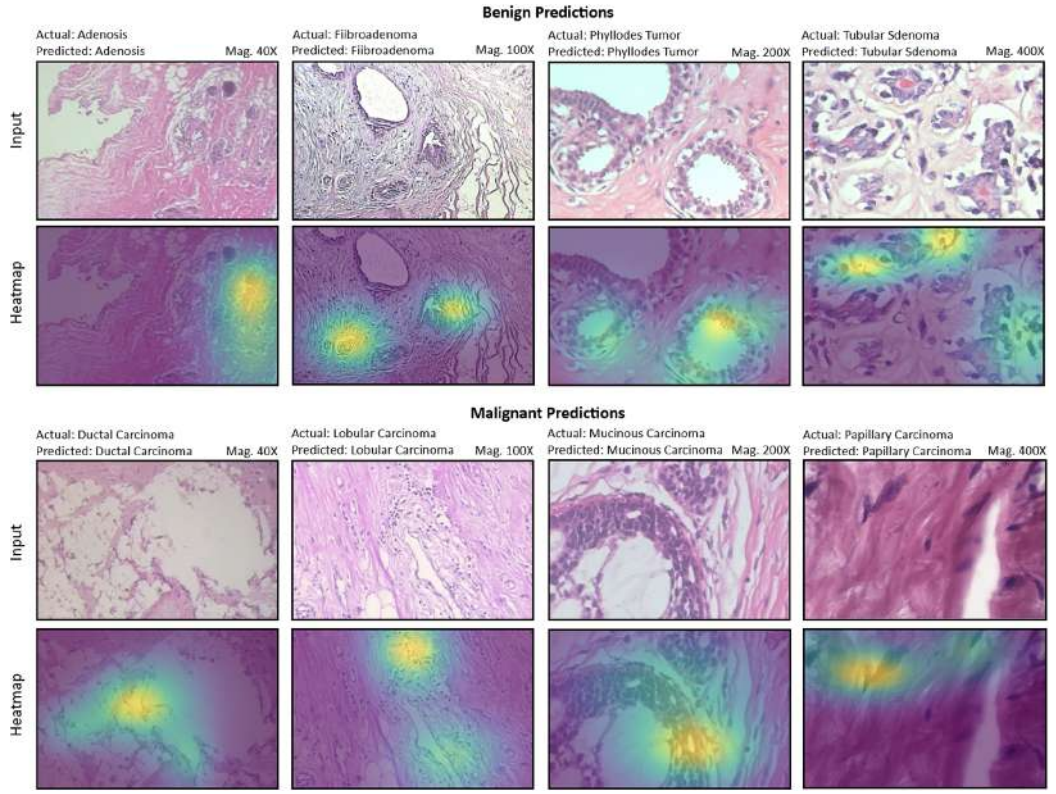


Fig. 7. Interpretability of the EVC-Net model demonstrated via Grad-CAM heatmaps on BreakHis dataset images. The top row showcases benign predictions (Actual/Predicted: Adenosis, Fibroadenoma, Phyllodes Tumor, Tubular Adenoma) and the bottom row malignant predictions (Actual/Predicted: Ductal Carcinoma, Lobular Carcinoma, Mucinous Carcinoma, Papillary Carcinoma), each with its corresponding input image and heatmap. Brighter colors (e.g., yellow) in the heatmaps indicate regions deemed most salient by EVC-Net for its classification decision, while darker colors (e.g., blue/purple) indicate less salient regions.

5.6 Computational Analysis and Practicality

Practical deployment of DL models necessitates an analysis of their computational cost. The EVC-Net framework, which integrates three distinct architectures, comprises approximately 21.5 million trainable parameters. All experiments were performed on a workstation equipped with an NVIDIA GeForce RTX 4090 GPU, establishing the hardware baseline for the following performance metrics.

The training process was performed using the 5-fold cross-validation scheme. With an Adam optimizer and a batch size of 128, the average training time was approximately 12 minutes per epoch. Full training for each fold, including early stopping, typically converged within 60–80 epochs, equating to a total of 12–16 hours of training time.

From a clinical deployment perspective, inference speed is the more critical metric. The EVC-Net model achieves an average inference time of 38 milliseconds per 224×224 image (approximately 26 frames per second). This speed is highly practical for a CAD system, as the clinical workflow does not require real-time video analysis. Rather, it demands the rapid classification of specific regions of interest or whole-slide image patches, for which a 38 milliseconds latency is more than sufficient

[12, 45]. The model's ability to provide a highly-accurate classification in milliseconds thus offers tangible, timely support for a pathologist's workflow.

While the hybrid nature of EVC-Net is more computationally intensive than a single-backbone model, it demonstrated high accuracy and stability (as shown in Tables 2–4) present a strong case for its utility. In clinical support systems where diagnostic precision is the primary objective, this tradeoff between computational cost and performance is well-justified.

5.7 Limitations and Future Work

Despite the promising performance of the EVC-Net framework, this study has several limitations that should be acknowledged. First, the model was developed and validated exclusively on the public BreakHis dataset. While BreakHis is a comprehensive and widely used benchmark, the model's generalization capabilities on unseen data from different clinical institutions, which may use different scanners or slide preparation protocols, have not yet been established. Second, the hybrid architecture of EVC-Net, which synergistically integrates an EfficientNetV2S, a ViT, and a capsule network, results in a model with significant computational complexity (approximately 21.5 million trainable parameters). This increased computational cost, compared to simpler, single-backbone models, could pose a practical challenge for deployment in resource-limited clinical environments.

Future work will aim to address these limitations. A critical next step is to evaluate EVC-Net's robustness and generalizability on diverse, multi-centric external datasets. Furthermore, exploring model compression and optimization techniques, such as pruning or quantization, could help reduce the computational footprint of EVC-Net, thereby improving its feasibility for real-world clinical adoption.

6 Conclusion

This article presented EVC-Net, a novel hybrid DL architecture for the classification of breast cancer from histopathological images, addressing both binary and multi-class (eight subtypes) tasks. The EVC-Net framework synergistically integrates an EfficientNetV2S for detailed local feature extraction, a ViT for global contextual information, and a capsule network to preserve spatial hierarchies within tissue structures. Extensive evaluation on the public BreakHis dataset demonstrated the model's robust performance, achieving an average binary classification accuracy of 0.985 and a multi-class accuracy of 0.954 across all magnifications. An ablation study further confirmed that each architectural component contributes effectively to the overall performance of the EVC-Net model. The empirical results indicate that EVC-Net provides an effective and robust approach for breast cancer histopathological image analysis. While acknowledging the study's limitations, such as validation on a single public dataset, the proposed framework offers a significant potential advancement for computer-aided diagnostic systems. Future work will focus on validating the EVC-Net model on diverse, multi-centric clinical datasets to further assess its generalizability and move it closer to practical application in clinical workflows.

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